

The Scott-T Transformer: Three-Phase to Two-Phase Conversion and its Simscape Implementation

Abstract

This note describes the Scott connection, which converts a three-phase system into a two-phase system whose two voltages have equal magnitude and are displaced by 90° . The connection rule and the $\sqrt{3}/2$ turns ratio of the teaser winding are derived, the current relations and the symmetrical-component behaviour under unbalanced two-phase load are given, and the equations implemented in the accompanying Simscape component `scott_t.ssc` are documented together with its per-unit parameterisation and sign conventions.

1 Introduction

A three-phase to two-phase transformation is required whenever a load or a source that is intrinsically two-phase, or a pair of single-phase loads that must be seen as balanced by the supply, has to be interfaced to a three-phase network. Typical applications are AC railway substations feeding two adjacent single-phase catenary sections, induction furnaces and electric arc furnaces, two-phase servo and control motors, and laboratory supplies for two-phase test benches.

The classical solution is the Scott connection, introduced by C. F. Scott at the end of the nineteenth century and still described in standard transformer texts [1, 2]. It uses two single-phase transformers, called the *main* and the *teaser*, and produces two secondary voltages of equal magnitude in exact quadrature. A functionally equivalent alternative, the Le Blanc connection, achieves the same result on a single three-limb core at the cost of a more complex winding arrangement [3].

The answer to the question that motivated this note is therefore affirmative: the two secondary phases of a Scott-connected transformer are displaced by exactly 90° , and this is a property of the connection geometry, not an approximation.

2 Connection

Figure 1 shows the arrangement. The main transformer primary, with N_1 turns, is connected between lines A and B and is centre-tapped at node O. The teaser transformer primary, with N_{1t} turns, is connected between line C and the centre tap O. Both secondaries have N_2 turns.

The essential feature is the centre tap. The teaser current returns through the two halves of the main primary in opposite winding senses, so the two contributions to the main-core magnetomotive force cancel. The two magnetic circuits are therefore decoupled even though the electrical circuits are not.

3 Voltage relations and the teaser turns ratio

Let the primary be supplied by a balanced positive-sequence system of phase voltages

$$\underline{V}_A = V\angle 0^\circ, \quad \underline{V}_B = V\angle -120^\circ, \quad \underline{V}_C = V\angle 120^\circ. \quad (1)$$

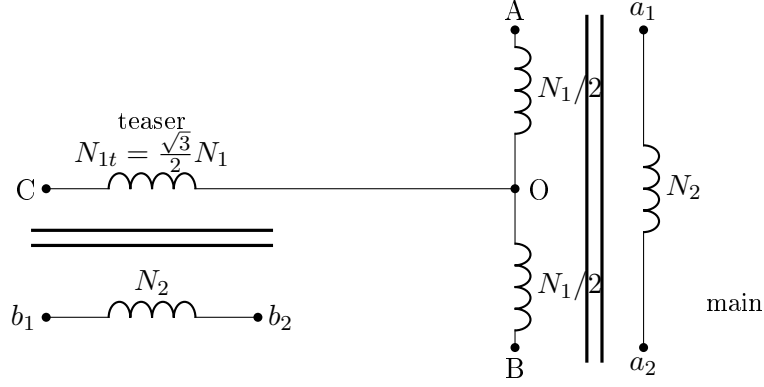


Figure 1: Scott connection. The teaser primary is returned to the centre tap of the main primary, so that its current splits equally between the two halves of the main winding and produces no net magneto-motive force in the main core.

Neglecting winding impedances, the voltage impressed on the main primary is

$$\underline{V}_{AB} = \underline{V}_A - \underline{V}_B = \sqrt{3} V \angle 30^\circ. \quad (2)$$

Because the two halves of the main primary are identical and carry the same induced voltage, the centre tap sits at the mean of the two line potentials,

$$\underline{V}_O = \frac{1}{2} (\underline{V}_A + \underline{V}_B) = -\frac{1}{2} \underline{V}_C, \quad (3)$$

where the last equality uses $\underline{V}_A + \underline{V}_B + \underline{V}_C = 0$. The voltage impressed on the teaser primary is therefore

$$\underline{V}_{CO} = \underline{V}_C - \underline{V}_O = \frac{3}{2} \underline{V}_C = \frac{3}{2} V \angle 120^\circ. \quad (4)$$

Two facts follow immediately. First, $\underline{V}_{CO}$ leads $\underline{V}_{AB}$ by exactly 90° : the quadrature is geometric and holds for any balanced supply. Second, the two primary voltages are *not* equal in magnitude, their ratio being $\frac{3}{2}V/\sqrt{3}V = \sqrt{3}/2$. To obtain equal secondary voltages with equal secondary turns, the teaser primary must have

$$\boxed{N_{1t} = \frac{\sqrt{3}}{2} N_1 \approx 0.8660 N_1} \quad (5)$$

Denoting $n = N_1/N_2$ and $V_1 = \sqrt{3}V$ the primary line-to-line rms voltage, the secondary voltages are

$$\underline{V}_a = \frac{\underline{V}_{AB}}{n} = \frac{V_1}{n} \angle 30^\circ, \quad \underline{V}_b = \frac{\underline{V}_{CO}}{N_{1t}/N_2} = \frac{V_1}{n} \angle 120^\circ. \quad (6)$$

The rated secondary phase voltage is thus $V_2 = V_1/n$ and $\underline{V}_b$ leads $\underline{V}_a$ by 90° . The phasor construction is shown in Fig. 2.

4 Current relations

Let $\underline{I}_a$ and $\underline{I}_b$ be the currents delivered by the main and teaser secondaries and let magnetizing currents be neglected. Magneto-motive force balance on the two cores gives

$$\frac{N_1}{2} (\underline{I}_{1a} + \underline{I}_{1b}) = N_2 \underline{I}_a, \quad N_{1t} \underline{I}_C = N_2 \underline{I}_b, \quad (7)$$

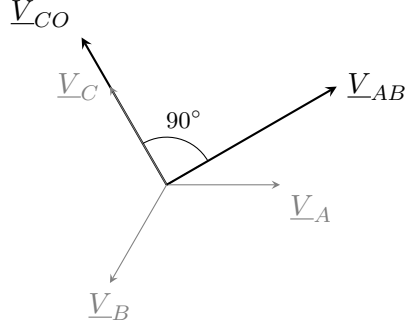


Figure 2: Phasor diagram. The main primary sees the line-to-line voltage $\underline{V}_{AB}$, the teaser primary sees $\frac{3}{2}\underline{V}_C$; the two are in quadrature and their magnitudes are in the ratio $2/\sqrt{3}$.

where $\underline{I}_{1a}$ is the current from A to O and $\underline{I}_{1b}$ the current from O to B, so that $\underline{I}_{1a} = \underline{I}_A$ and $\underline{I}_{1b} = -\underline{I}_B$. With Kirchhoff's current law at node O, $\underline{I}_A + \underline{I}_B + \underline{I}_C = 0$, the line currents are

$$\underline{I}_A = \frac{1}{n} \left(\underline{I}_a - \frac{1}{\sqrt{3}} \underline{I}_b \right), \quad \underline{I}_B = \frac{1}{n} \left(-\underline{I}_a - \frac{1}{\sqrt{3}} \underline{I}_b \right), \quad \underline{I}_C = \frac{2}{\sqrt{3}n} \underline{I}_b. \quad (8)$$

For a balanced two-phase load, $\underline{I}_b = j\underline{I}_a$, and (8) reduces to a balanced three-phase set of magnitude $2|\underline{I}_a|/(\sqrt{3}n)$.

Behaviour under unbalanced two-phase load

Applying the symmetrical-component transformation to (8) yields the compact result

$$\underline{I}_+ = \frac{e^{-j30^\circ}}{\sqrt{3}n} (\underline{I}_a - j\underline{I}_b), \quad \underline{I}_- = \frac{e^{+j30^\circ}}{\sqrt{3}n} (\underline{I}_a + j\underline{I}_b). \quad (9)$$

The negative-sequence current vanishes if and only if $\underline{I}_b = j\underline{I}_a$, that is, only for a two-phase load that is balanced both in magnitude and in phase angle. The Scott connection does not by itself compensate an unbalanced load; it only maps a balanced two-phase load onto a balanced three-phase supply. In traction substations this is the reason why the two catenary sections must be loaded symmetrically, or why additional compensation is required [3].

Rating

Each transformer carries one half of the total apparent power, $S_r/2$. The main transformer is rated for V_1 on the primary and the teaser for $(\sqrt{3}/2)V_1$, both for the same secondary voltage V_2 . The two units are therefore not interchangeable. In practice the two windings are often made identical up to the tap position, and both are provided with the $\sqrt{3}/2 \approx 0.866$ tap plus the centre tap, so that either unit can act as main or as teaser.

5 Simscape model

The component `scott_t.ssc` implements the connection as a single Simscape block with seven external electrical nodes: A, B, C on the primary side and a1, a2, b1, b2 on the secondary side. The centre tap 0 is an internal private node. The two secondaries are galvanically separate; tying a2 to b2 externally produces a two-phase three-wire system with a common return.

5.1 Equations

Winding resistances and leakage inductances are lumped in series with each winding, and each core is represented by a linear magnetizing inductance in parallel with a core-loss resistance, both referred to the secondary. Denoting by e_m and e_t the secondary induced electromotive forces, the implemented equations are

$$v_A - v_O = R_{1h} i_{1a} + L_{1h} \frac{di_{1a}}{dt} + \frac{n}{2} e_m, \quad (10)$$

$$v_O - v_B = R_{1h} i_{1b} + L_{1h} \frac{di_{1b}}{dt} + \frac{n}{2} e_m, \quad (11)$$

$$v_C - v_O = R_{1t} i_{1t} + L_{1t} \frac{di_{1t}}{dt} + n_t e_t, \quad (12)$$

$$v_{a1} - v_{a2} = e_m - R_2 i_{2m} - L_2 \frac{di_{2m}}{dt}, \quad (13)$$

$$v_{b1} - v_{b2} = e_t - R_2 i_{2t} - L_2 \frac{di_{2t}}{dt}, \quad (14)$$

with the magneto-motive force balance referred to the secondary windings,

$$\frac{n}{2} (i_{1a} + i_{1b}) - i_{2m} = i_{Lm} + \frac{e_m}{R_{fe}}, \quad n_t i_{1t} - i_{2t} = i_{Lt} + \frac{e_t}{R_{fe}}, \quad (15)$$

and the magnetizing branches

$$e_m = L_m \frac{di_{Lm}}{dt}, \quad e_t = L_m \frac{di_{Lt}}{dt}. \quad (16)$$

Here $n = N_1/N_2 = V_1/V_2$ and $n_t = N_{1t}/N_2 = (\sqrt{3}/2)n$. Kirchhoff's current law at the internal node, $i_{1a} + i_{1t} = i_{1b}$, is enforced automatically by the branch declarations.

Two points are worth noting. First, the main primary is split into two halves of $N_1/2$ turns each, with half the total resistance and half the total leakage inductance; the two halves are not mutually coupled through their leakage paths, which is the usual approximation. Second, the sum $i_{1a} + i_{1b}$ drives the main core while their difference, equal to the teaser return current, does not. The centre-tap cancellation is thus a property of the equations and does not need to be imposed separately.

5.2 Parameters

The impedances are entered in per unit. Each transformer processes $S_r/2$, so the secondary base impedance is

$$Z_{b2} = \frac{V_2^2}{S_r/2} = \frac{2V_2^2}{S_r}, \quad (17)$$

and the primary bases follow from the respective turns ratios, $Z_{b1,\text{main}} = n^2 Z_{b2}$ and $Z_{b1,\text{teaser}} = n_t^2 Z_{b2} = \frac{3}{4} n^2 Z_{b2}$. With this choice the same per-unit value applies to both units. Table 1 lists the parameters.

5.3 Sign conventions

Positive current enters the component at **A**, **B** and **C** and leaves it at **a1** and **b1**. With a balanced positive-sequence supply of line-to-line rms voltage **V1** applied with phase sequence A–B–C, the open-circuit secondary voltages are

$$v_{a1} - v_{a2} = \sqrt{2} V_2 \cos(\omega t + 30^\circ), \quad v_{b1} - v_{b2} = \sqrt{2} V_2 \cos(\omega t + 120^\circ), \quad (18)$$

so the teaser secondary leads the main secondary by 90° . Reversing the phase sequence at the primary reverses the sign of the quadrature, which is the expected behaviour and a useful check.

Table 1: Parameters of the `scott_t` component.

Name	Description	Unit	Default
<code>Sr</code>	Rated apparent power, three-phase total	V A	100×10^3
<code>V1</code>	Primary rated line-to-line voltage, rms	V	400
<code>V2</code>	Secondary rated phase voltage, rms	V	230
<code>fr</code>	Rated frequency	Hz	50
<code>r1</code>	Primary winding resistance	pu	0.005
<code>l1</code>	Primary leakage inductance	pu	0.030
<code>r2</code>	Secondary winding resistance	pu	0.005
<code>l2</code>	Secondary leakage inductance	pu	0.030
<code>lm</code>	Magnetizing inductance	pu	100
<code>rfe</code>	Core-loss resistance	pu	500

5.4 Limitations

The magnetic model is linear: saturation, hysteresis and frequency-dependent core losses are not represented, and no thermal port is provided. Inrush phenomena are therefore not reproduced. Leakage coupling between the two halves of the main primary is neglected, which slightly underestimates the impedance seen by the teaser return current. If saturation is required, the two magnetizing relations $e = L_m di_L/dt$ can be replaced by a flux-linkage formulation $e = d\lambda/dt$ with a nonlinear $\lambda(i_L)$ characteristic, at the cost of one additional differential variable per core.

6 Suggested verification

A minimal test model consists of a balanced three-phase voltage source connected to `A`, `B`, `C`, with `a2` tied to `b2` and a resistive load on each secondary. Three checks are sufficient:

1. *No load.* The two secondary voltages must be equal in rms value and in exact quadrature, and the magnetizing currents must be in the ratio 1 : 1 referred to the secondary.
2. *Balanced load.* With equal resistive loads on both secondaries, the three line currents must form a balanced positive-sequence set; by (9) the negative-sequence component must be zero to within the magnetizing current.
3. *Single-phase load.* Loading only the main secondary must produce a negative-sequence line current of magnitude equal to the positive-sequence one, in agreement with (9) for $\underline{I}_b = 0$.

References

- [1] A. E. Fitzgerald, C. Kingsley Jr., and S. D. Umans, *Electric Machinery*, 6th ed. New York, NY, USA: McGraw-Hill, 2003.
- [2] J. J. Winders Jr., *Power Transformer Principles and Applications*. New York, NY, USA: Marcel Dekker, 2002.
- [3] B.-K. Chen and B.-S. Guo, "Three-phase models of specially connected transformers," *IEEE Trans. Power Del.*, vol. 11, no. 1, pp. 323–330, Jan. 1996.
- [4] *IEEE Standard for General Requirements for Liquid-Immersed Distribution, Power, and Regulating Transformers*, IEEE Std C57.12.00.
- [5] The MathWorks, Inc., *Simscape Language Guide*, Natick, MA, USA.